

# CprE 381, Computer Organization and Assembly-Level Programming

## Lab 2 Report

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**Submit a typeset pdf version of this on Canvas by the due date. Refer to the highlighted language in the lab document for the context of the following questions.**

[Part 3 (a)] Draw the interface description (i.e., the “symbol” or high-level blackbox) for the RISC-V register file. Which ports do you think are necessary, and how wide (in bits) do they need to be?

[INSERT IMAGE]

If you read this, I am almost finished with the lab but haven't filled this out yet. Don't worry. I know you said fill it out while you do it, but it's hard to draw on waveforms on Linux, so I have to do that on Windows and it's faster to do them all at once after everything is done.

*A complete register file requires:*

*Required for writing:*

- 32-bit input bus (which can set the value of a register in the register file)
- 5-bit write address bus (5 bits can specify one of 32 registers) to select a register to write to
- Write enable line that determines whether the value in the input should be written to the register specified by the write address bus.

*Required for reading:*

- 32-bit output bus A that carries the output of one register
- 32-bit output bus B that carries the output of another register
- 5-bit read address bus A to specify what address the output bus A uses.
- 5-bit read address bus B to specify what address the output bus B uses.

[Part 3 (b)] Create an N-bit register using this flip-flop as your basis.  
Complete.

[Part 3 (c)] Waveform.

[Part 3 (d)] What type of decoder would be required by the RISC-V register file and why?

[Part 3 (e)] Waveform.

[Part 3 (f)] In your write-up, describe and defend the design you intend on implementing for the next part.

[Part 3 (g)] Waveform.

[Part 3 (h)] Draw a (simplified) schematic (i.e., components within the high-level blackbox) for the RISC-V register file, using the same top-level interface ports as in your solution describe above and using only the register, decoder, and mux VHDL components you have created.

[Part 3 (i)] Waveform.

[Part 4 (b)] Draw a symbol for this RISC-V-like datapath.

[Part 4 (c)] Draw a schematic of the simplified RISC-V processor datapath consisting only of the component described in part (a) and the register file from problem (1).

[Part 4 (d)] Include in your report waveform screenshots that demonstrate your properly functioning design. Annotate what the final state of the register file should be and provide a mapping of the registers used in your drawing to the register names.

[Part 5 (a)] Read through the mem.vhd file, and based on your understanding of the VHDL implementation, provide a 2-3 sentence description of each of the individual ports (both generic and regular).

[Part 5 (c)] Waveforms and hex file.

[Part 6 (a)] What are the RISC-V instructions that require some value to be sign extended? What are the RISC-V instructions that require some value to be zero extended? What are the RISC-V instructions that require some value to be appended with zero?

[Part 6 (b)] what are the different “extender” components that would be required by a RISC-V processor implementation (i.e., number of bits of input and sign vs zero extension vs zero appending)?

[Part 6 (d)] Waveform.

[Part 7 (a)] what control signals will need to be added to the simple processor from part 2? How do these control signals correspond to the ports on the mem.vhd component analyzed in part 3?

[Part 7 (b)] Draw a schematic of a simplified RISC-V processor consisting only of the base components used in part 2, the extender component described in part 4, and the data memory from part 3.

[Part 7 (c)] Waveform.

[Part 7 (d)] Compare the Questasim waveform/memory list against the RARs output/memory list.